1.5 Dividing Multi-Digit Numbers With Decimals to the Thousandths

Lesson Introduction

Benchmark	MA.6.NSO.2.1 Multiply and divide positive multi-digit numbers with decimals to the thousandths, including using a standard algorithm with procedural fluency.	Learning Target	• I can fluently divide multi-digit numbers with decimals to the thousandths.

	Building On:	Working Toward:
Benchmark Coherence	MA.5.NSO.2.5 Multiply and divide a multi-digit number with decimals to the tenths by one-tenth and one-hundredth with procedural reliability.	MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.

Essential Questions	• How do you divide decimals using long division? • Why does multiplying the dividend and divisor by the same power of 10 not change the value of the quotient?
Lesson Narrative	The lesson begins with a Warm-Up and Refresher that reviews long division from 5th grade and operations with decimals. The lesson then continues by exploring the relationship between the dividend, divisor, and quotient in long division with decimals. The lesson is designed for students to build understanding of the size of the quotient in terms of the dividend and divisor – not to teach shortcuts about decimal placement.

Component Summary	1.5.1 Warm-Up (~5 min.)	"Notice and Wonder" on decimal quotients *(SE p. 26)*	1.5.4 Bring It Together (5-10 min.)	Fill-in-the-blank activity to check conceptual understanding *(SE p. 28)*
	1.5.2 Refresher (~5 min.)	Review of long division with whole numbers *(SE p. 27)*	1.5.5 Guided Practice (5-10 min.)	Practice dividing decimals and recognizing how to multiply decimal values before dividing *(SE p. 29)*
	1.5.3 Exploration (15-20 min.)	Explore strategies for dividing decimals with values multiplied by powers of 10 *(SE p. 27)*	1.5.6 Your Turn! (5-10 min.)	Practice dividing decimals using long division *(SE p. 29)*
	1.5.7 Wrap-Up (~5 min.)	Reflection: Self-assessment of learning targets *(SE p. 30)*		

	Glossary	Materials	Additional Preparation
Resources	Key Terms: N/A Additional Terms: • Quotient • Dividend • Divisor • Powers of 10	• (Optional) Chart paper • (Optional) Markers	N/A

Teacher Tips

Common Misconceptions	Things to Consider
• Students may struggle with placement of the decimal in the quotient. • Students may confuse steps of the standard algorithm for long division.	• This lesson is built to make connections from students' prior knowledge of long division to now include decimals. Consider making explicit connections and highlighting how the decimal placement is determined. • To reinforce correct placement of the decimal in the quotient, consider building students' understanding of base-10 thinking using similar thinking as modeled in 1.5.3 Exploration.

Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
Notice and Wonder 1.5.1 Warm-Up positions students to notice similarities and differences between values of quotients when dividing by decimals. Consider applying "Notice and Wonder" routines to encourage students' math thinking and help students to think about the size of the quotient in terms of the dividend and divisor.	**MA.K12.MTR.3.1 Mathematical Fluency** Throughout the lesson, students will be working to divide multi-digit decimals confidently and fluently to the thousandths place using long division. Consider using the 1.5.6 Your Turn! section to provide practice with generalizing how to multiply by powers of 10 to have a whole number divisor.

Differentiate Instruction

1.5.6 Your Turn! – Consider specifying which problems students complete by difficulty based on student confidence and comfort with long division.

1.5.3 Exploration and 1.5.5 Guided Practice - Consider allowing partners to discuss reasoning and check each other's work on procedural accuracy with the long division algorithm. This partner work can also help auditory and visual learners hear and see different strategies and correct minor long division mistakes.

1.5.5 Guided Practice - Consider having students create a math journal entry or interactive notebook page to solve each problem clearly with long division. Consider creating an anchor poster on chart paper and/or using this work as a visual reference for correct long division with decimals for students to refer to in later lessons.

Lesson Notes:

1.5.1 Warm-Up: Quotients as Decimals

Component Overview: *Students use "Notice and Wonder" routines to look at quotients of several division expressions, all including 1 as the dividend and differing values as the divisor. Consider having students share in partners or as a class. Highlight responses that indicate how the size of the quotient changes in relation to the divisor's value.*

1.5.1 Warm-Up: Quotients as Decimals

The quotients of several division expressions are shown.

$1 \div 2 = 0.5$	$1 \div 3 = 0.333\overline{3}$	$1 \div 4 = 0.25$
$1 \div 5 = 0.2$	$1 \div 6 = 0.666\overline{6}$	$1 \div 7 = 0.\overline{142857}$
$1 \div 8 = 0.125$	$1 \div 9 = 0.111\overline{1}$	$1 \div 10 = 0.1$
$1 \div 11 = 0.0909\overline{09}$	$1 \div 12 = 0.08\overline{33}$	$1 \div 13 = 0.\overline{076923}$

1. Write something you notice about the values of the quotients and something you still wonder.

I notice...	I wonder...
Answers will vary. I notice that the repeating decimals all have odd divisors. I notice that the quotients decrease as the divisor increases.	*Answers will vary. I wonder if there is a way to predict repeating decimals.*

Notice and Wonder
Ask the students to share what they noticed and wondered. Consider highlighting responses that focus on the size of the quotient changing in relation to the divisor as a primer for the lesson.

1.5.2 Refresher: Long Division

Component Overview: *Students review the long division algorithm by filling in incomplete steps on a partially completed problem. Use this section as an opportunity to review procedural steps to divide using the long division algorithm and highlight important steps.*

1.5.2 Refresher: Long Division

Jiang was dividing using long division and got stuck.

1. Complete the problem to show Jiang the missing steps.

$$
\begin{array}{r|rrrr}
 & & 1 & \boxed{2} & \boxed{5} \\
\hline
15 & 1 & 8 & 7 & 5 \\
 & -1 & 5 & & \\
\cline{2-5}
 & & \boxed{3} & \boxed{7} & \\
 & - & \boxed{3} & \boxed{0} & \\
\cline{3-5}
 & & & \boxed{7} & \boxed{5} \\
 & & - & \boxed{7} & \boxed{5} \\
\cline{4-5}
 & & & & 0
\end{array}
$$

2. What advice about long division do you have that could help Jiang in the future?
   Answers will vary.

Consider the purpose of this lesson when gauging time in this activity. Emphasize question 2 and the relationship between the divisor and dividend in relation to quotient, and utilize that same line of thinking when engaging with decimals.

Consider asking students to discuss this question rather than writing their answers. This entry point might engage auditory learners. Use sentence stems to encourage conversation such as, "One step Jiang needs to remember in the future is…" or "One detail that I have to remember when I divide is…"

1.5.3 Exploration: Dividing Decimals

Component Overview: *Students work in partners to explore two strategies for finding the quotient of a decimal dividend and whole number divisor. Use this section to help students notice that multiplying the dividend and divisor by the same power of 10 does not change the value of the quotient, and to pay attention to the similarities and differences in the placement of the digits/decimal in the long division algorithm. Consider providing opportunities for students to share thinking and check long division work with partners.*

1.5.3 Exploration: Dividing Decimals

Giovanny and Alina used different strategies to find the quotient of $48.78 \div 9$.

Giovanny's Strategy

```
      5 ¦ 4  2
 9 ) 4 8 ¦ 7  8
   − 4 5 ¦
       3 ¦ 7
   −   3 ¦ 6
         ¦ 1  8
       − ¦ 1  8
         ¦    0
```

Alina's Strategy

```
              5 ¦ 4  2
 9 0 0 ) 4 8 7 8 ¦
       − 4 5 0 0 ¦
           3 7 8 ¦ 0
       −   3 6 0 ¦ 0
             1 8 ¦ 0  0
       −     1 8 ¦ 0  0
                0 ¦    0
```

$48.78 \div 9 = 5.42$

Work with your partner to answer the questions about Giovanny and Alina's strategies.

1. How are the two students' calculations alike and different?

Similarities	Differences
• *Same quotient* • *Same non-zero digits in dividend and divisor*	• *Giovanny's strategy keeps the values as decimals* • *Alina multiplies the dividend and divisor by* 100 *first*

Notice and Wonder

Prompt students to notice how the strategies and digits used are alike and different, specifically regarding the placement of digits and how their placement affects place value. Encourage discussion using sentence stems like "I notice the digit in Giovanny's strategy is…and the same digit in Alina's strategy is…" This attention to digits and place value will build conceptual understanding of decimal long division.

While circulating the room, monitor for students who are stuck or unsure what to look for when comparing.

- *What is the same or different about the values Giovanny and Alina used as their divisor and dividend?*
- *What is the same or different about the quotient?*
- *What does the vertical line represent?*
- *How does the vertical line change in both strategies and what does this mean in relation to place value?*

1.5.3 Exploration: Dividing Decimals (continued)

2. Look at Giovanny's calculations. Complete the statements with the appropriate place value to explain how you can tell that the 36 means "36 tenths" and the 18 means "18 hundredths" in his calculations.

   - The 3 in 36 is in the ___*ones*___ place and the 6 is in the ___*tenths*___ place, so 36 in this calculation means 36 tenths.

   - The 1 in 18 is in the ___*tenths*___ place, so it has the same value as 10 hundredths. The 8 is in the ___*hundredths*___ place. Together, they make 18 hundredths.

3. Look at Alina's calculations. Complete the statement to describe the values of 3600 and 1800 in her calculation.

   - In Alina's calculations, the 3600 represents 3600 ___*tenths*___, and the 1800 represents 1800 ___*hundredths*___.

4. Why is the value of $48.78 \div 9$ the same as the value of $4878 \div 900$?
   The numbers in the second expression are both multiplied by 100, so the value of the quotient will not change.

5. Explain why $51.84 \div 6.4$ has the same value as $518.4 \div 64$.
   The numbers in the second expression are both multiplied by 10, so the value of the quotient will remain the same.

6. Find the value of $51.84 \div 6.4$ using long division. Show your work.

$$\begin{array}{r} 8.1 \\ 6.4\overline{)51.84} \\ -\,51.2 \\ \hline 0.64 \\ -\,0.64 \\ \hline 0 \end{array}$$

Encourage students to complete the statements while also verbally explaining their reasoning as an entry point for auditory learners.

Consider probing questions to encourage synthesis for students as they formalize their own approach to dividing with decimals:

- *What are some advantages of calculating the quotient with the decimals intact, like Giovanny?*
- *What are some advantages of rewriting the dividend and divisor as whole numbers before dividing, like Alina?*
- *Why does multiplying the dividend and divisor by the same power of 10 not change the value of the quotient?*

1.5.4 Bring It Together: Dividing Decimals

Component Overview: *Students complete statements to confirm conceptual understanding of why the value of a quotient does not change when both the divisor and dividend are multiplied by the same power of 10. Consider allowing students to complete the statements individually first, then check for correctness and add complexity by having students provide an example as evidence.*

1.5.4 Bring It Together: Dividing Decimals

1. Complete the summary of the exploration of dividing decimals using the words and phrases from the bank below.

does not change	multiply
power of 10	whole numbers

One way to find a quotient of two decimals is to first

___*multiply*___ each decimal by a ___*power of* 10___ so

that both values are ___*whole numbers*___. If we multiply both

values by the same power of 10, this ___*does not change*___

the value of the quotient.

Consider adding complexity by having students write their own example of two division statements whose quotient is the same that have different decimal divisors and dividends. Students can write their examples in the space below and have a partner confirm their division statements are equivalent.

1.5.5 Guided Practice: Dividing Decimals

Component Overview: *Students practice dividing decimals by first multiplying the divisor and dividend by a power of 10. Consider emphasizing conceptual understanding and procedural fluency by having students solve problems using various versions of the divisor and dividend that will have the same quotient.*

1.5.5 Guided Practice: Dividing Decimals

1. Find each quotient using a strategy similar to Giovanny's or Alina's. Show your work.

   a. $24.2 \div 1.1$
   22

   b. $17.028 \div 0.08$
   212.85

Consider probing questions for students who are stuck that align with the benchmark for this lesson:

- *If you need a whole number divisor before dividing, how do you determine which power of 10 to multiply both decimals by?*
- *Which power of 10 will you use for question 1a? For question 1b? Why?*

2. Coby and Olivia each found the quotient of $10.92 \div 2.6$. The values they each used are listed.

   - Coby calculated $1092 \div 260$.
   - Olivia calculated $109.2 \div 26$.

   a. Will the values they used result in the same quotient as $10.92 \div 2.6$? Explain.
   Yes, Coby multiplied both values by 100 first and Olivia multiplied both values by 10 first.

   b. Find the quotient of $10.92 \div 2.6$.
   4.2

Notice the skills and higher order thinking necessary for engaging students in question 2. While monitoring during this practice time, consider highlighting two differing responses from students and have them share their thinking with the class.

1.5.6 Your Turn!

Component Overview: *Students apply learning in this section by dividing decimals using any strategy. Consider choosing which problems students work on based on learner preparedness and using a math journal or interactive notebook page to record strategies as a reference for students in later lessons.*

1.5.6 Your Turn!

1. Find each quotient. Show your work.

   a. $43.396 \div 57.1$
   0.76

   b. $29.88 \div 0.36$
   83

   c. $262.5 \div 5.25$
   50

   d. $15.129 \div 12.3$
   1.23

Consider using specific problems from this section for different learner fluency with divisibility rules. For example, questions 1a and 1d include multiplying by 1,000 and therefore requires students to think about divisibility with larger values. Questions 1b and 1c involve multiplying by 100 and therefore includes smaller whole values to divide.

1.5.7 Wrap-Up: Reflection

Component Overview: *This self-assessment asks students to consider their procedural understanding and comfort with the lesson's learning target. Encourage students to revisit work throughout the lesson as evidence of learning.*

1.5.7 Wrap-Up: Reflection

1. Draw an **X** on the line to indicate where your current learning is in relation to the learning target.
   Answers will vary.

   I can fluently divide multi-digit numbers with decimals to the thousandths.

I need help. I can't get started.	I'm getting there.	I understand and have evidence.

2. Briefly explain why you placed the X where you did.
   Answers will vary.

Consider pairing the "number" line responses seen here with informal observations throughout the lesson for intentional grouping in Lesson 7 that targets division of fractions.